

Course Schedule

Week	Date	Topics	Tools
Week 1	Tue 19 May 2026	Foundations of Data • Sets, Relations, Tuples • Keys • Basic SQL (SELECT, WHERE) • Logic-first thinking	SQLite (VS Code)
Week 2	Tue 26 May 2026	Relational Modelling • JOINS • Subqueries • CTEs (non-recursive) • Normalisation (1NF/2NF)	PostgreSQL (Standalone installation / Docker + VS Code)
Week 3	Tue 2 June 2026	Analytical Modelling • Kimball • Fact/Dimension • SCD1/SCD2 • CASE expressions • Indexes (conceptual)	PostgreSQL
Week 4	Tue 9 June 2026	NoSQL Thinking • Document Modelling • Embedding vs Referencing • MongoDB Aggregation Pipeline	MongoDB (Standalone installation / Docker)
Week 5	Tue 16 June 2026	Graph Thinking • Recursive CTEs • Adjacency Lists • Apache AGE • Cypher Patterns • AI-ready Graph Reasoning	PostgreSQL (Apache AGE optional)
Week 6	Tue 23 June 2026	ACID • Transactions • Stored Procedures • Spark DataFrames • Lakehouse Concepts • ETL on Databricks	PostgreSQL + Databricks (Cloud service)

Course Material

All weekly materials will be released on the course GitHub repository:

https://github.com/lyh1068/DMS_SMM695

For installation instructions and environment setup, visit:

https://github.com/lyh1068/DMS_SMM695/blob/main/environment/ Please make sure you have the required tools installed before each lesson.

I will update Moodle regularly once my access is activated. In the meantime, please continue to use GitHub as the primary source for course materials.

Contact

The best way to reach me is by email: Yuanheng.Li.2@[City University email domain]

(Domain intentionally omitted here for security.)

I will make every effort to respond to emails by the end of the week in which they are received.

Final Course Project

Students will work in groups of four, forming a total of eleven groups (with one group of five). The assignment will be released in the third lesson, and the submission deadline is **8 July**.

Deliverables include:

- code
- documentation
- a presentation slide deck

I'm also open to your ideas — this project can become a strong addition to your CV. If you have a suggestion or direction you'd like to explore, please email me before **5 June**. Thoughtful suggestions may earn **1–2 bonus points**.